

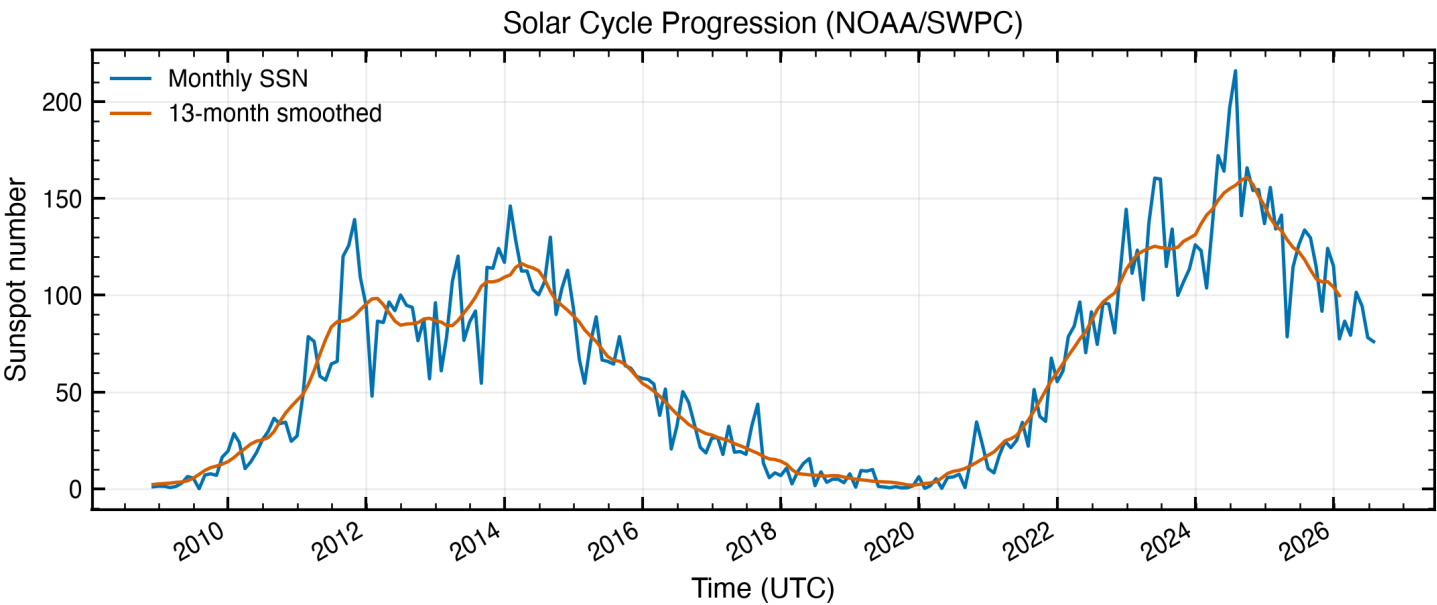
Reporting window and provenance

Window: 2026-09-01T11:00:00 to 2026-09-02T11:00:00 UTC. Sources: GOES primary XRS 1-min, NOAA RTSW solar wind at L1, NOAA planetary Kp, NOAA/SWPC solar region analysis, NASA CCMC DONKI, NOAA Solar Cycle Progression. Operational feeds - not science quality; every number is audit-logged (workspace/logs/audit.jsonl):

fetch_swpc_timeseries: ok (4baeadde69f5)
find_flares: ok (463901bd02c6)
fetch_swpc_timeseries: ok (12950ffb1d1c)
fetch_swpc_timeseries: ok (c2f9d5cc1cf8)
fetch_swpc_timeseries: ok (611cc42f043c)
get_solar_regions: ok (f37aa8513a0c)
search_donki: ok (d767fd3b7b79)
fetch_solar_cycle: ok (55baaac31ca9)
describe_series: ok (262f80ccf24f)
describe_series: ok (cdd3592e8272)
plot_timeseries: ok (ef42461459c4)
plot_stack: ok (a4179fcfc49b)
plot_timeseries: ok (a8f4611ffd4f)

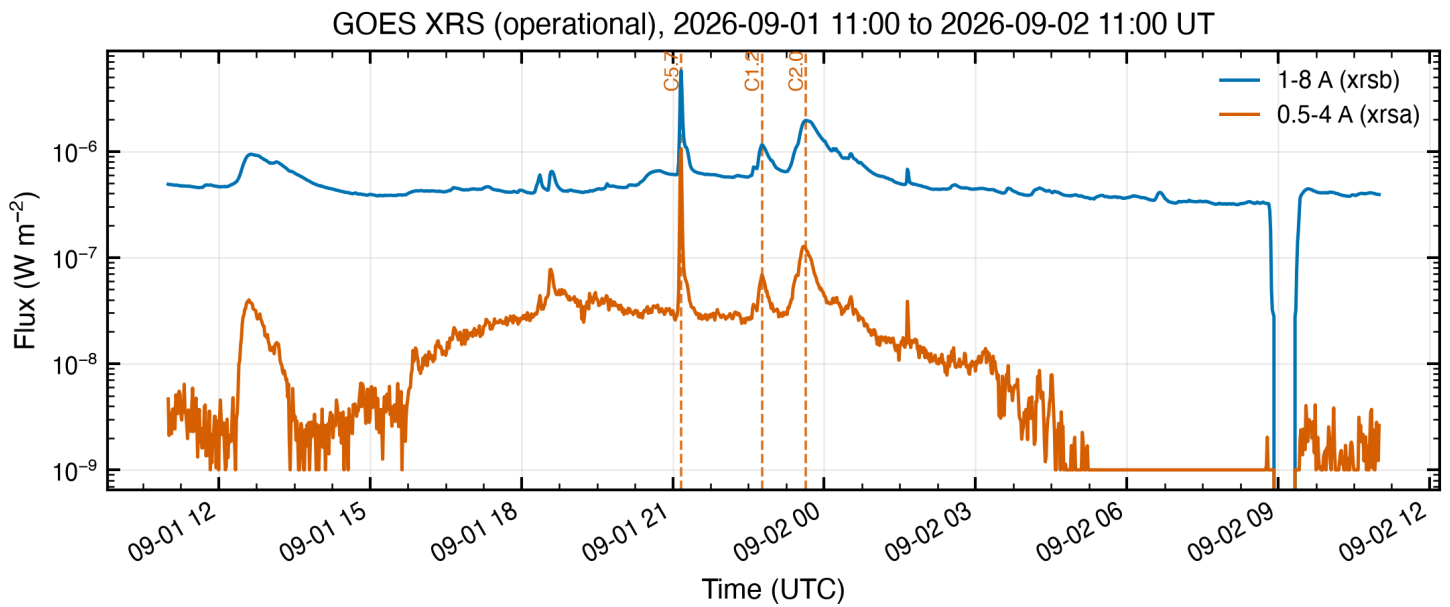
Solar cycle

Latest monthly sunspot number: 76.0 (2026-08; previous month 78.1). Cycle 25 smoothed maximum 160.9 in 2024-10 (smoothed series final through 2026-02).



Flare activity

Detected 0 X, 0 M, 3 C flares (plus 33 B-level enhancements; sensitive detector - the official SWPC event list applies stricter criteria). Strongest: C5.7 (peak 21:10 UT, 118 min).



Sunspot regions (NOAA analysis)

Analysis date 2026-09-02:

AR4520 S12W33 mag B 4 spots, area 40 mu

AR4523 N10E49 mag A 2 spots, area 50 mu

AR4521 N09E39 mag A 1 spots, area 80 mu

CMEs (DONKI)

2026-09-01T06:00:00-CME-001: Faint CME observed to the SE in SOHO LASCO C2 and one frame of STEREO A COR2, fading into background and weakly visible in LASCO C3 and GOES CCOR-1. T

2026-09-01T21:24:00-CME-001: CME first seen to the East by SOHO LASCO C2 beginning at 2026-09-01T21:24Z, as well as by SOHO LASCO C3 and GOES CCOR-1 in later frames. This CME is n

2026-09-01T23:00:00-CME-001: CME first seen to the NE by SOHO LASCO C2 beginning at 2026-09-01T23:00Z, as well as by SOHO LASCO C3 and GOES CCOR-1 in later frames. This CME is not

2026-09-02T10:24:00-CME-001: Very faint CME first seen to the E in SOHO LASCO C2 starting around 2026-09-02T10:24Z. It is more easily seen to the E in STEREO A COR2. It is not see

Solar wind and geomagnetic conditions

Proton speed 360-609 km/s (mean 389); density mean 3.7 /cm³. Bt 1.0-7.0 nT; Bz GSM -5.5 to 5.5 nT. Kp 0.67-2.67.

Solar wind at L1 (RTSW) + Kp, 2026-09-01/2026-09-02

